# Model Evaluation Report: Balanced vs. Imbalanced Training

## 1. Summary of Model Performance

| Model | Test Set | Accuracy | Key Observations |
|---|---|---|---|
| Model_A (Balanced) | Balanced Test Set | 0.473 | Reasonable performance across all classes; |
| Model_A (Balanced) | Imbalanced Test Set | 0.569 | Performs well on majority class (5*), lower on |
| Model_B (Imbalanced) | Balanced Test Set | 0.285 | Bias towards 5*; poor performance on minorit |
| Model_B (Imbalanced) | Imbalanced Test Set | 0.689 | Performs well on majority class (5*); minority |

## 2. Observations

Balanced Model (Model_A):

- Maintains better fairness across all ratings.
- Accuracy drops slightly when applied to imbalanced test set because it hasn't seen as many majority-class reviews.

Imbalanced Model (Model_B):

- Strong performance on majority class (5*) in imbalanced test set.
- Poor generalization on balanced test set; overfits to the majority class.

Effect of training distribution:

- Balanced training ensures consistent performance across all classes.
- Imbalanced training favors majority class at the expense of minority classes.

## 3. Recommendation

- Deploy Model_A (Balanced) if fairness across ratings is important (real-world scenario where all ratings matter).
- Use Model_B (Imbalanced) only if most reviews are expected to be 5* and the goal is to optimize for majority-class accuracy.